Myeloproliferative neoplasms — PV, ET, PMF

Philadelphia-negative MPNs driven by constitutive JAK-STAT signaling (**JAK2 V617F, CALR, MPL**) · goals: **prevent thrombosis** in PV/ET, **control spleen, symptoms and anemia** in MF, and **transplant** the fit high-risk MF patient

No Stanford Oncology Handbook chapter — compiled from WHO 2022 / ICC 2022 criteria · NCCN MPN · ELN · primary trial reports verified against PubMed; approvals against FDA labels, October 2026.

DRIVER MUTATIONS

Driver	PV	ET	PMF
JAK2 V617F *exon 14*	~95%	50–60%	50–60%
JAK2 exon 12	~3%	—	—
CALR *exon 9*	—	20–25%	25–30%
MPL *W515L/K*	—	3–5%	5–8%
Triple-negative	—	10–15%	~10%

JAK2-negative erythrocytosis is not PV until exon 12 is excluded — go back to the secondary differential. **CALR never occurs in PV** (mutant CALR signals only through MPL). In ET, CALR = higher platelets but **lower thrombosis risk** than JAK2; in PMF, **CALR type 1 is favorable** and triple-negative is worst.

DIAGNOSIS (WHO 2022 · ICC 2022)

Entity	Criteria
PV	**Hb >16.5 (M) / >16.0 (F) or Hct >49% / >48%** + trilineage marrow hypercellularity with pleomorphic mature megakaryocytes + **JAK2 V617F or exon 12**. Minor: **subnormal EPO**. All 3 major, or first 2 + minor. Marrow can be skipped if Hb >18.5 / >16.5 (Hct >55.5 / >49.5%) with JAK2 + low EPO — but it misses early fibrosis
ET	**Platelets ≥450** + marrow with large, hyperlobulated (“staghorn”) megakaryocytes, **no left shift, reticulin ≤grade 1** + not another myeloid neoplasm + **JAK2** / **CALR** / **MPL**. Minor: other clonal marker or no reactive cause. **Marrow is required**
Prefibrotic PMF	Megakaryocyte proliferation **with atypia** (clustering, cloud-like nuclei), **reticulin ≤grade 1**, hypercellular, granulocytic ↑ / erythroid ↓ + driver or clonal marker + ≥1 minor: anemia, WBC ≥11, palpable spleen, ↑LDH
Overt PMF	Same megakaryocyte atypia + **reticulin/collagen grade 2–3** + clonal marker + ≥1 minor (above, or **leukoerythroblastosis**)
Post-PV / ET MF	Prior PV/ET + **grade 2–3 fibrosis** + ≥2 of: anemia (post-PV: or loss of phlebotomy/cytoreduction need; post-ET: Hb ↓≥2 g/dL), leukoerythroblastosis, enlarging spleen (≥5 cm), constitutional symptoms, ↑LDH (post-ET) (IWG-MRT)

Prefibrotic PMF vs ET — same CBC, but prefibrotic PMF has worse survival and more fibrotic/leukemic progression. It turns on **megakaryocyte morphology**, so every suspected ET needs a marrow.

TREATMENT PATHWAY

Polycythemia vera

All patients
phlebotomy to Hct <45% (CYTO-PV)
aspirin 81 mg daily; treat CV risk factors
no iron supplementation

↓

Low risk (<60, no thrombosis)
phlebotomy + aspirin
cytoreduce if phlebotomy-intolerant, symptomatic, rising WBC/plt or splenomegaly: **ropeg** (Low-PV)

↓

High risk (≥60 or prior thrombosis)
ropeginterferon or **hydroxyurea**
HU-resistant/intolerant: **ruxolitinib** or IFN
phlebotomy-dependent: add **rusfertide** (FDA 2026)

Essential thrombocythemia

Very low / low risk
observe, or aspirin 81 mg if JAK2+ or CV RFs
CALR+, no CV RF: skip aspirin (bleeding)
plt >1000: check vWF:RCo first

↓

Intermediate (>60, JAK2 WT)
aspirin 81 mg daily
cytoreduce if symptoms or extreme platelets

↓

High (thrombosis, or >60 + JAK2+)
aspirin + **hydroxyurea** (PT-1)
or **ropeginterferon** (FDA any line; SURPASS-ET 2L)
or anagrelide; IFN if young / pregnancy

Myelofibrosis (PMF, post-PV/ET)

Risk-stratify → transplant decision
DIPSS-plus / MIPSS70+ v2 / MYSEC-PM
int-2 / high risk, fit: refer for allo-HCT
low risk, asymptomatic: observe

↓

Symptoms or spleen — JAK inhibitor
plt ≥50 + Hb OK: **ruxolitinib** (OS signal)
Hb <10 / transfusions: **momelotinib**
plt <50: **pacritinib**
after ruxolitinib: fedratinib, momelotinib

↓

Anemia / blast phase
anemia: ESA (EPO <500), danazol, momelotinib
BP (≥20% blasts): HMA ± ven → allo-HCT

WORKUP

- **Peripheral blood JAK2 V617F** (qPCR, report VAF) → if negative: **CALR + MPL** (thrombocytosis / fibrosis) or **JAK2 exon 12** (erythrocytosis). Most labs now run an **MPN NGS panel** up front
- **NGS for high-molecular-risk mutations**: **ASXL1, SRSF2, U2AF1 Q157, EZH2, IDH1/2** (+ TP53, RAS) — drive MIPSS70+ and transplant referral
- **BCR::ABL1** (PCR or FISH) in every thrombocytosis / leukocytosis — CML can present as isolated thrombocytosis
- **Erythrocytosis: serum EPO** (low → PV; normal/high → secondary), ferritin/iron (iron deficiency masks PV), SpO_2, sleep study, smoking/CO-Hb, **testosterone/SGLT2i**, renal/hepatic mass imaging
- **Thrombocytosis:** exclude iron deficiency, inflammation/infection, asplenia, malignancy, bleeding (CRP, ferritin)
- **Marrow aspirate + biopsy** with reticulin stain + karyotype; abdominal US for spleen; LDH, uric acid
- **Platelets >1000–1500 or any bleeding: vWF:RCo (ristocetin cofactor) + vWF antigen** — acquired vWD; ratio <0.7 or RCo <30% → hold aspirin, cytoreduce

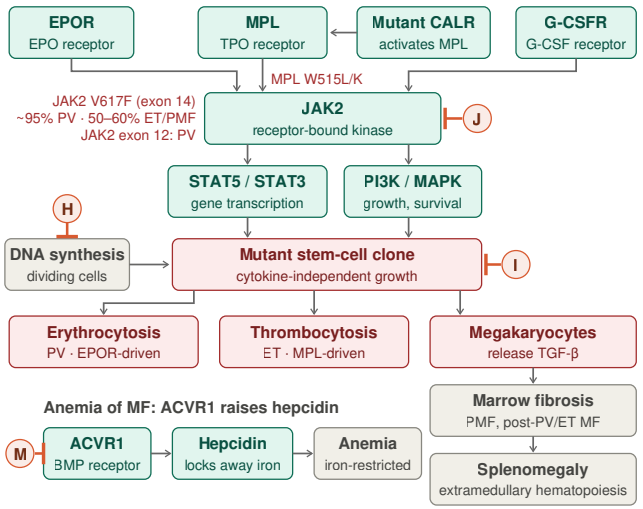
RISK STRATIFICATION

Score	Factors · groups
PV	**High = age ≥60 OR prior thrombosis**; low = neither. Leukocytosis, CV risk factors and JAK2 VAF add risk but are not in the score
ET — revised IPSET-thrombosis	**Very low:** ≤60, no thrombosis, JAK2 WT · **low:** ≤60, no thrombosis, JAK2+ · **intermediate:** >60, no thrombosis, JAK2 WT · **high:** prior thrombosis, or >60 + JAK2+
DIPSS-plus *PMF, any time*	DIPSS (age >65, Hb <10 [2 pts], WBC >25, blasts ≥1%, constitutional sx) + **unfavorable karyotype, plt <100, transfusion need**. Median OS: low 185 · int-1 78 · int-2 35 · high 16 mo
MIPSS70+ v2.0 *PMF, molecular*	Very-high-risk karyotype 4, unfavorable karyotype 3, **≥2 HMR mutations 3** / 1 HMR 2, **absent CALR type 1** 2, constitutional sx 2, severe anemia 2 (moderate 1), blasts ≥2% 1 → very low / low / intermediate / high / very high
MYSEC-PM *post-PV/ET MF*	Hb <11 (2), blasts ≥3% (2), **CALR-unmutated** (2), plt <150 (1), constitutional sx (1), +0.15/yr of age. Median OS: low NR · int-1 9.3 · int-2 4.4 · high 2 yr

Use MYSEC-PM, not DIPSS/IPSS, for secondary MF. HMR = ASXL1, SRSF2, U2AF1 Q157, EZH2, IDH1/2; **TP53** (multi-hit) predicts leukemic transformation.

Myeloproliferative neoplasms — PV, ET, PMF

PATHOPHYSIOLOGY



(J) JAK inhibitors: ruxolitinib (JAK1/2) · fedratinib (JAK2) · pacritinib (JAK2/IRAK1) · momelotinib (JAK1/2)

(M) momelotinib: ACVR1 inhibitor (H) hydroxyurea: ribonucleotide reductase inhibitor

(I) ropeginterferon · peginterferon: interferon-α, preferentially targets mutant stem cells

Hydroxyurea lowers counts non-selectively; JAK inhibitors reduce spleen and symptoms; neither changes allele burden much. Interferon can lower JAK2/CALR allele burden.

FOLLOW-UP AND PROGRESSION

- **Stable PV/ET:** CBC every 3–6 mo, symptom score (MPN-SAF TSS), spleen exam, CV risk factors; reassess risk category with each new thrombosis or birthday past 60
- **Suspect post-PV/ET MF:** falling Hb or loss of phlebotomy need, rising LDH, leukoerythroblastosis, growing spleen, new constitutional symptoms → **repeat marrow + NGS**, restage with MYSEC-PM
- **Suspect accelerated/blast phase:** circulating blasts ≥10%, new refractory cytopenias → marrow, karyotype, NGS (TP53), HLA typing
- **JAK2/CALR VAF** is worth tracking on interferon (molecular response); not needed on HU

OTHER MPNS (BRIEF)

- **MPN-U** — MPN features that fit no category (often early PMF or splanchnic thrombosis presentations); manage by the dominant phenotype
- **CNL** — sustained neutrophilia ≥25 (≥80% segs/bands), no dysplasia; **CSF3R T618I** (~60–80%); ruxolitinib responses; allo-HCT for fit
- **CEL** — clonal eosinophilia ≥1.5 with marrow abnormality; first exclude **PDGFRA/B, FGFR1, JAK2 rearrangements** (imatinib-sensitive PDGFR)

POLYCYTHEMIA VERA — EVIDENCE

Question	Answer · trial
Hct target	**<45% for everyone** (CYTO-PV, n=365): CV death or major thrombosis **2.7% vs 9.8%** at 45–50% (HR 3.91)
Aspirin	81–100 mg daily for all without contraindication (ECLAP: fewer thromboses)
Interferon vs HU	**PROUD-PV / CONTINUATION-PV** (ropeg vs HU, early PV): 12-mo non-inferiority **not met** (CHR + normal spleen 21% vs 28%), but CHR at 36 mo **71% vs 51%** and deeper JAK2 VAF falls. **Responses accrue slowly**
Low-risk PV	**Low-PV** (randomized phase 2, age ≤60): ropeg 100 µg q2wk + phlebotomy kept Hct ≤45% in **84% vs 60%**; stopped early for efficacy
HU-resistant / intolerant	**Ruxolitinib** — RESPONSE (phase 3): Hct control + spleen response **21% vs 1%**; MAJIC-PV (phase 2): CR **43% vs 26%**, **event-free survival HR 0.58**
Phlebotomy-dependent	**Rusfertide** (hepcidin mimetic, VERIFY phase 3, n=293): no phlebotomy wk 20–32 **76.9% vs 32.9%**; Hct <45% 62.6% vs 14.4%. **FDA 28 Aug 2026** (Mimrylo). Erythrocytosis only — **not cytoreductive**

HU resistance (ELN): need for phlebotomy, plt >400 + WBC >10, or no spleen response after 3 mo at ≥2 g/day; cytopenia at lowest effective dose; or intolerable toxicity (leg ulcers, mucositis).

ESSENTIAL THROMBOCYTHEMIA — EVIDENCE

Question	Answer · trial
Aspirin by risk	JAK2+ or CV risk factors: **aspirin 81 mg**. **Low-risk CALR+**: aspirin did not cut thrombosis but **raised bleeding** (retrospective) — observe. BID aspirin is an option if JAK2+ with CV risk factors (faster platelet turnover)
HU vs anagrelide	**PT-1** (n=809, high risk, + aspirin): anagrelide arm **more arterial thrombosis, serious bleeding and myelofibrosis**, fewer VTE (composite OR 1.57) → **HU first line**
Second line	**Ropeginterferon** — SURPASS-ET (phase 3, HU-resistant/intolerant + WBC >10): durable ELN response **43% vs 6%** vs anagrelide; thromboembolic events 1 vs 10 (FDA label). **FDA 31 Aug 2026** (all adult ET). 96% Asian cohort
Platelet goal	Cytoreduce to symptom control and **plt <400–450**; no evidence that normalizing counts alone prevents thrombosis in low-risk disease

MYELOFIBROSIS — JAK INHIBITOR CHOICE

Agent	Use · evidence
Ruxolitinib *JAK1/2*	**Default first line, plt ≥50.** COMFORT-I: SVR35 **41.9% vs 0.7%**, TSS50 46% vs 5%, deaths HR 0.50; COMFORT-II vs BAT: SVR35 28% vs 0%. Worsens anemia/plt early
Fedratinib *JAK2*	JAKARTA (1L): SVR35 36% vs 1%. **FREEDOM2 (post-rux): 36% vs 6%** vs BAT. **Thiamine** (Wernicke in 4 at 500 mg)
Pacritinib *JAK2/IRAK1*	**Plt <50** (accelerated 2022). PERSIST-2 (plt ≤100): 200 mg BID SVR35 **22% vs 3%**, TSS50 32% vs 14%. Confirmatory PACIFICA (plt <50) not yet reported
Momelotinib *JAK1/2 + ACVR1*	**Anemia (Hb <10)**, FDA 2023. MOMENTUM (JAKi-exposed, anemic): TSS50 **25% vs 9%** vs danazol + better transfusion independence. SIMPLIFY-1: SVR35 non-inferior to rux (26.5 vs 29%); symptoms not

Add-on combinations: spleen up, symptoms not, none approved. With ruxolitinib in JAKi-naive MF, SVR35 roughly doubled with **pelabresib** (MANIFEST-2: 66% vs 35%), **navitoclax** (TRANSFORM-1: 63% vs 32%), **selinexor** (SENTRY: 50% vs 28%) — symptom endpoints not met in all three; pelabresib has an **excess leukemic transformation signal**; US filing deferred.

ANEMIA, TRANSPLANT, BLAST PHASE

- **Anemia of MF: ESA** if EPO <500 and not transfusion-dependent; **danazol** (monitor LFTs, PSA); **switch to momelotinib**. **Luspatercept** (INDEPENDENCE, transfusion-dependent on JAKi): RBC-TI 23% vs 13%, **primary endpoint not met** (P=0.067) — off-label
- **Allo-HCT** — only curative therapy. Refer **DIPSS int-2/high, MIPSS70+ high, HMR mutations, transfusion dependence, plt <100, TP53/ complex karyotype**, age generally <70. Continue JAKi up to conditioning (taper over days)
- **MTSS** (post-transplant): age ≥57, KPS <90, plt <150, WBC >25, HLA-mismatched unrelated donor, **ASXL1**, non-CALR/MPL driver → 5-yr OS **83 / 64 / 37 / 22%**
- **Accelerated (10–19% blasts) / blast phase (≥20%):** median OS ~**3–6 mo**. **HMA + venetoclax** gives CR/CRi ~40–45% but median OS only ~7–8 mo (retrospective series) — use as a **bridge to allo-HCT**, the only route to long-term survival. Intensive induction gives no OS edge without transplant; IDH-mutant: ivosidenib/enasidenib

Myeloproliferative neoplasms — PV, ET, PMF

AGENTS — DOSING

Agent	Dosing
Aspirin	**81 mg daily**; 81 mg BID option in JAK2+ ET with CV risk factors. Hold if vWF:RCo <30%
Phlebotomy	250–500 mL to **Hct <45%**; check CBC q4–6 wk once stable. No iron
Hydroxyurea	**500 mg BID** (15–20 mg/kg/day) → titrate to counts; avoid in pregnancy
Ropeginterferon alfa-2b *Besremi*	**PV:** 100 µg SC q2wk (50 if on HU) → +50 µg q2wk to max **500 µg** until counts stable. **ET: 250 → 350 (wk 2) → 500 µg q2wk** from wk 4
Peg-IFN alfa-2a *off-label*	45–90 µg SC weekly, titrate to 180 µg; **pregnancy-compatible**
Rusfertide *Mimrylo*	**19 mg SC weekly** → titrate (9.5–108 mg) to hold Hct <45%; CBC q2–4 wk during titration
Anagrelide	Label: 0.5 mg QID or 1 mg BID (0.5 mg BID often used) → ↑ ≤0.5 mg/day per week; max 10 mg/day, 2.5 mg/dose. **ECG** + cardiac history (QT, palpitations, HF)
Ruxolitinib *MF*	By platelets: **>200 → 20 mg BID** · 100–200 → 15 BID · 50–<100 → 5 BID; titrate. **PV: 10 mg BID**
Fedratinib	**400 mg daily** (plt ≥50); 200 mg if CrCl 15–29 or strong CYP3A4 inhibitor. **Thiamine level at baseline; replete before start**; prophylactic antiemetic
Pacritinib	**200 mg BID** (plt <50); avoid with QTc >480, active bleeding
Momelotinib	**200 mg daily**; no platelet floor on the label (MOMENTUM enrolled plt ≥25); dose-reduce for cytopenias per label
Danazol	200 mg TID (600 mg/day); LFTs, PSA in men
ESA	Epoetin alfa 40,000 U weekly or darbepoetin 150–300 µg q2wk if EPO <500; stop if no response by 12 wk

RESPONSE DEFINITIONS

Disease	Response
PV *ELN CHR*	Hct <45% **without phlebotomy**, plt ≤400, WBC ≤10, normal spleen, no disease symptoms
ET *ELN CHR*	Plt ≤400, WBC <10, normal spleen, no symptoms, no thrombosis/bleeding
MF *trial endpoints*	**SVR35** = ≥35% spleen volume reduction on MRI/CT at wk 24 · **TSS50** = ≥50% fall in MFSAF total symptom score · anemia: **transfusion independence** ≥12 wk

TOXICITY AND MONITORING

- **Hydroxyurea:** macrocytosis (expected), cytopenias, **leg / oral ulcers**, **NMSC** (annual skin exam). Leukemogenic risk is low and not proven
- **Interferons: boxed warning** — depression/suicidality, autoimmune thyroiditis, retinopathy, hepatotoxicity. Baseline + periodic **TSH, LFTs, eye exam**, mood screen; avoid with uncontrolled psychiatric or autoimmune disease
- **JAK inhibitors:** anemia/thrombocytopenia (nadir wk 8–12), **herpes zoster** (recombinant zoster vaccine), **HBV reactivation** (screen HBsAg/anti-HBc), TB, PML, **NMSC**, lipids ↑. MACE/VTE/malignancy class warnings from tofacitinib in RA
- **Ruxolitinib discontinuation syndrome:** abrupt stop → symptom/spleen rebound, rarely SIRS-like collapse — **taper**, steroids if needed
- **Fedratinib: Wernicke encephalopathy** — thiamine monthly early, stop for confusion/ataxia/vision change and give IV thiamine; GI toxicity, amylase/lipase
- **Rusfertide:** injection-site reactions (56%), anemia (16%) per label; **may worsen thrombocytosis** — monitor platelets, may need cytoreduction; induced iron restriction is intended — **don't give iron**

SPECIAL SITUATIONS

- **Splanchnic vein thrombosis** / **Budd-Chiari: test JAK2 V617F** (then CALR) even with normal counts — portal hypertension masks erythrocytosis and thrombocytosis. Long-term anticoagulation + cytoreduction (IFN or HU)
- **Pregnancy:** aspirin throughout; **peg-IFN** if cytoreduction needed (stop HU, anagrelide, JAKi before conception); Hct <45% in PV; **LMWH for 6 wk postpartum**, antepartum too if prior thrombosis
- **Surgery:** control counts first (Hct <45%, plt near normal); hold aspirin 7 days for high-bleed procedures; aggressive VTE prophylaxis — perioperative thrombosis and bleeding both rise
- **Extreme thrombocytosis** (>1000–1500): acquired vWD → bleeding, not clotting; check vWF:RCo, hold aspirin if low, cytoreduce. Plateletpheresis only for acute thrombosis/bleeding
- **Pruritus** (aquagenic, PV): antihistamines, SSRIs (paroxetine), narrowband UVB; **IFN or ruxolitinib** for refractory itch
- **Thrombosis on therapy:** arterial → aspirin + cytoreduce; venous → DOAC/warfarin long term + cytoreduce; reassess Hct/plt control

KEY TRIALS

Trial	Design · result
CYTO-PV *2013*	Hct <45% vs 45–50%: CV death/thrombosis **2.7% vs 9.8%**
PROUD / **CONTINUATION-PV** *2020*	Ropeg vs HU: 12-mo NI missed; CHR 36 mo **71% vs 51%**
Low-PV *2021, ph 2*	Ropeg + phlebotomy in low-risk PV: response **84% vs 60%**
RESPONSE *2015*	Rux vs BAT, HU-R/I PV: composite **21% vs 1%**
MAJIC-PV *2023, ph 2*	Rux vs BAT: CR 43% vs 26%; **EFS HR 0.58**
VERIFY *2025*	Rusfertide vs placebo: no phlebotomy **77% vs 33%**
PT-1 *2005*	HU vs anagrelide in high-risk ET: **HU superior** (OR 1.57)
SURPASS-ET *2025*	Ropeg vs anagrelide, 2L ET: response **43% vs 6%**
COMFORT-I / **II** *2012*	Rux: SVR35 **42% vs 1%** (placebo); 28% vs 0% (BAT)
JAKARTA / **FREEDOM2**	Fedratinib SVR35 36% vs 1% (1L); **36% vs 6%** post-rux
PERSIST-2 *2018*	Pacritinib, plt ≤100: SVR35 **18% vs 3%**
MOMENTUM *2023*	Momelotinib vs danazol, anemic: TSS50 **25% vs 9%**
SIMPLIFY-1 *2017*	Momelotinib vs rux: SVR35 non-inferior; TSS not